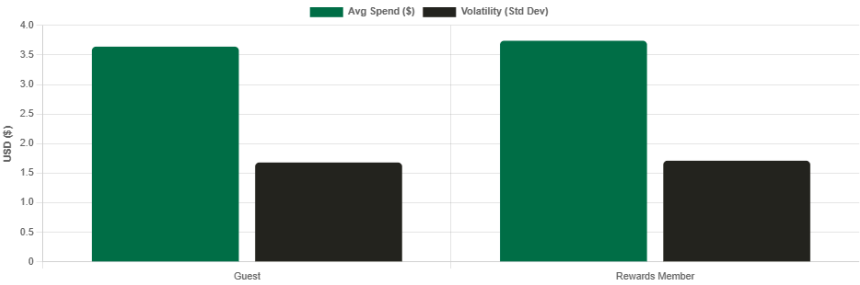


Starbucks Data Insights & Query Audit

Consumer Behavior: Spending & Volatility



Strategic Inference

The analysis shows that **Rewards Members** (~\$3.75) and **Guests** (~\$3.65) exhibit nearly identical spending patterns. The **Volatility (Standard Deviation)** remains consistent across both groups. This indicates the loyalty program has not yet established a more predictable or "clustered" high-spending habit compared to non-members.

Audit Metric	Technical Finding
Row Count Variance	EXPLAIN estimated ~99,119 rows, while ANALYZE confirmed 100,000. This 1% variance indicates highly accurate table statistics, ensuring the optimizer makes reliable path decisions.
Bottleneck Identification	The Aggregate using temporary table remains the most expensive step, consuming ~790ms. While indexing speeds up data retrieval, complex math (AVG/STDDEV) still requires significant CPU resources.
Efficiency Risks	The presence of Using temporary and Using filesort warns that the database is creating on-the-fly disk storage for grouping, which is a major performance risk as data scales.

Optimization Impact Walkthrough

Baseline (Unoptimized)

```
-> Table scan on distribution_spending (actual time=1128..1128 rows=2 loops=1)
-> Materialize CTE distribution_spending (actual time=1128..1128 rows=2 loops=1)
-> Table scan on <temporary> (actual time=1125..1125 rows=2 loops=1)
```

Optimized (With Index)

```
-> Table scan on distribution_spending (actual time=791..791 rows=2 loops=1)
-> Materialize CTE distribution_spending (actual time=791..791 rows=2 loops=1)
-> Nested loop left join (actual time=0.422..725 rows=100000 loops=1)
```

- ✔ **STEP IMPROVED:** Nested Loop Join / Table Access
- ⚡ **PERFORMANCE GAIN:** Execution time dropped from 1,128ms to 791ms (~30% faster).
- 🔍 **RATIONALE:** Shifted from 'ALL' (Full Table Scan) to 'ref' (Index Lookup). The database now jumps directly to matching IDs.

